

Assignment 10: Data Scraping

Lexi Nelson

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
#load needed packages
library(tidyverse)
library(here);
here() #pointing me to the Assignments folder where I am currently working
```

```
## [1] "/home/guest/ENV872/EDE_Spring2026/Assignments"
```

```
library(rvest)
#deleted previous code loading lubridate and dplyr
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Set the year to 2024
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
#indicate the NC Water site URL to be scraped
webpage <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024')
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- Contact Person
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
#scrape the water system name
water_system_name <- webpage %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()
water_system_name
```

```
## [1] "Durham"
```

```
#scrape the contact person
contact_person <- webpage %>%
  html_nodes("div+ table tr:nth-child(4) td:nth-child(2)") %>%
  html_text()
contact_person
```

```
## [1] "Kieu Tran"
```

```
#scrape the ownership
ownership <- webpage %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()
ownership
```

```
## [1] "Municipality"
```

```
#scrape maximum day use for each month
```

```
MGD_month <- webpage %>%  
  html_nodes("th~ td+ td") %>%  
  html_text()  
MGD_month
```

```
## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"  
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4  
#convert scraped data into a dataframe  
df_withdrawals <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),  
                             "WaterSystemName" = rep(water_system_name, 12),  
                             "ContactPerson" = rep(contact_person, 12),  
                             "Ownership" = rep(ownership, 12),  
                             "MaximumDailyWithdrawal" = as.numeric(MGD_month),  
                             "Year" = rep(2024, 12))
```

```
#add date column with month and year in data format  
#I am not sure what data format means. this comes from instructions.  
#I will assume Date format
```

```
df_withdrawals <- df_withdrawals %>%  
  mutate(Date = as.Date(paste(Year, Month, "01", sep = "-")))  
class(df_withdrawals$Date) #make sure it is still in date format
```

```
## [1] "Date"
```

```
#show(df_withdrawals) avoid displaying dataframes in knit output
```

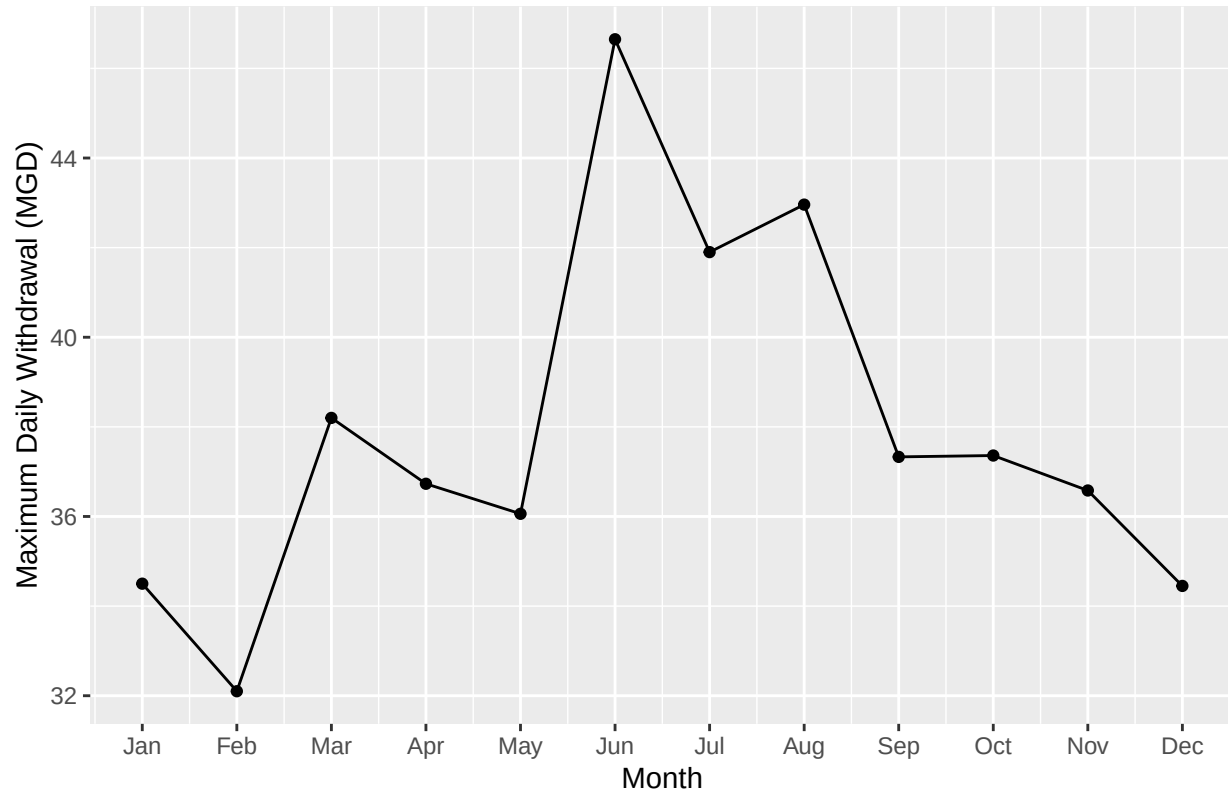
```
#5  
#create line plot of maximum daily withdrawals across months  
df_withdrawals_plot <-  
  ggplot(df_withdrawals, aes(x = Month, y = MaximumDailyWithdrawal)) +  
  geom_line() +  
  geom_point() +
```

```

scale_x_continuous(breaks = 1:12, labels = month.abb) +
#making sure months show on x axis since I got points off for that last hw
labs(
  title = "Maximum Daily Withdrawals Across Months in 2024",
  x = "Month",
  y = "Maximum Daily Withdrawal (MGD)"
)
show(df_withdrawals_plot)

```

Maximum Daily Withdrawals Across Months in 2024



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```

#6.
#construct a scraping function
scrape.it <- function(the_year, the_PWSID){
  #Get the proper url
  the_url <-
    paste0('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=', the_PWSID,

```

```

    '&year=',the_year)
print(the_url)

#Fetch the website
the_website <- read_html(the_url)

#Scrape the data
#<enter code from above here>
the_water_system_name <- the_website %>%
  html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>% html_text()
the_contact_person <- the_website %>%
  html_nodes('div+ table tr:nth-child(4) td:nth-child(2)') %>% html_text()
the_ownership <- the_website %>%
  html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>% html_text()
the_MGD_month <- the_website %>% html_nodes('th~ td+ td') %>% html_text()

#Convert to dataframe
#<enter code from above here, adding a column for the month>
the_df <- data.frame(
  Month = c(1,5,9,2,6,10,3,7,11,4,8,12),
  Water_System_Name = rep(the_water_system_name, 12),
  Contact_Person = rep(the_contact_person, 12),
  Ownership = rep(the_ownership, 12),
  Maximum_Daily-Withdrawal = as.numeric(the_MGD_month),
  Year = rep(the_year, 12))

#Return the dataframe
return(the_df)
}

# test the above function
df2024 <- scrape.it(2024, "03-32-010")

## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024"

#show(df2024)

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7
#use the function to pull max daily withdrawals in 2020
df2020 <- scrape.it(2020, "03-32-010")

## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2020"

#show(df2020)
#show the structure of the dataframe
glimpse(df2020)

```

```
## Rows: 12
## Columns: 6
## $ Month                <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ Water_System_Name    <chr> "Durham", "Durham", "Durham", "Durham", "Durh~
## $ Contact_Person       <chr> "Kieu Tran", "Kieu Tran", "Kieu Tran", "Kieu ~
## $ Ownership            <chr> "Municipality", "Municipality", "Municipality~
## $ Maximum_Daily-Withdrawal <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37.~
## $ Year                 <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 202~
```

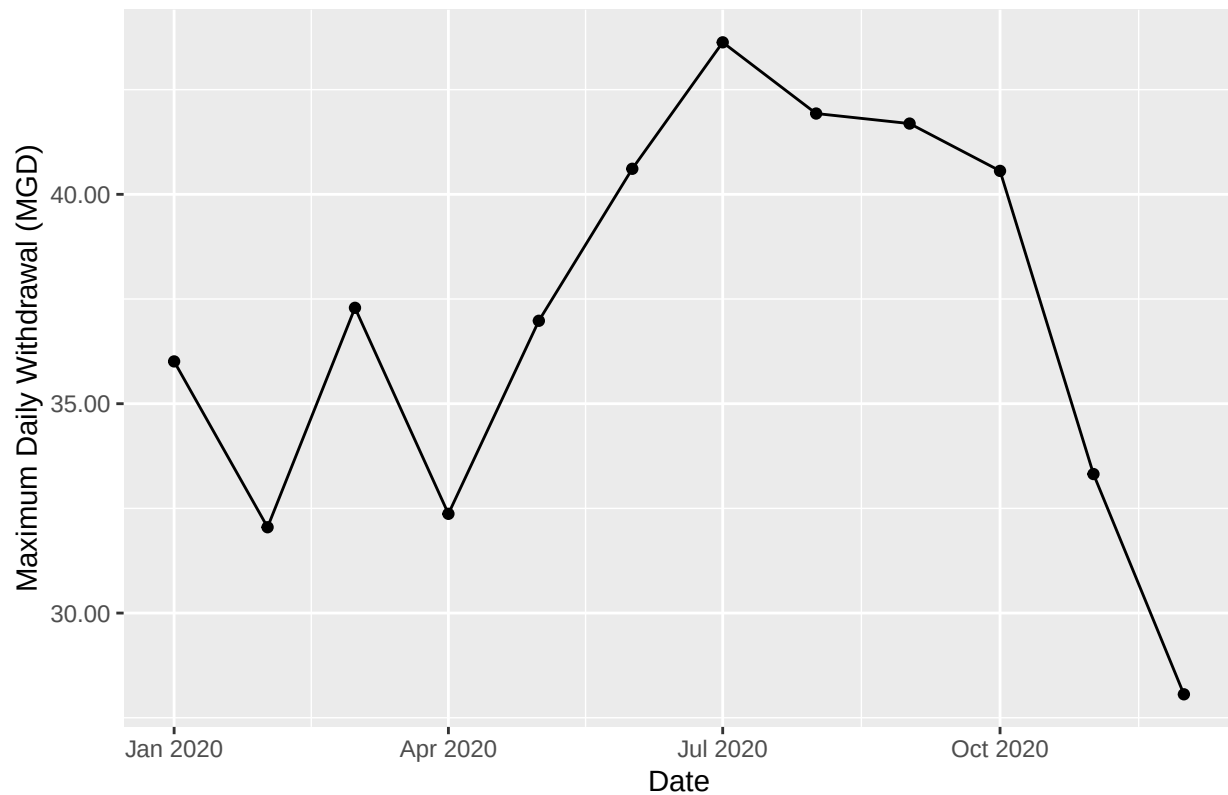
```
#make sure MGD_month is numeric to avoid errors down the line
```

```
df2020$Maximum_Daily-Withdrawal <- as.numeric(df2020$Maximum_Daily-Withdrawal)
df2020$Date <- as.Date(sprintf("%d-%02d-01",
                                df2020$Year, df2020$Month))
```

```
#code to plot max daily withdrawals
```

```
df_withdrawals_2020 <-
  ggplot(df2020, aes(x = Date, y = Maximum_Daily-Withdrawal, group=1)) +
  geom_line() +
  geom_point() +
  scale_x_date(date_labels = "%b %Y") +
  #revised from scale_x_continuous
  scale_y_continuous(labels = scales::label_number(accuracy = 0.01)) +
  #making sure months show on x axis since I got points off for that last hw
  labs(
    title = "Maximum Daily Withdrawals Across Months for Durham",
    x = "Date",
    y = "Maximum Daily Withdrawal (MGD)"
  )
show(df_withdrawals_2020)
```

Maximum Daily Withdrawals Across Months for Durham



8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
#use function to extract data for Cary
dfC2020 <- scrape.it(2020, "03-92-020")
```

```
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2020"
```

```
#show(dfC2020)
#show the structure of the dataframe
glimpse(dfC2020)
```

```
## Rows: 12
## Columns: 6
## $ Month                <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ Water_System_Name    <chr> "Cary", "Cary", "Cary", "Cary", "Cary", "Cary~
## $ Contact_Person       <chr> "Jamie Revels", "Jamie Revels", "Jamie Revels~
## $ Ownership            <chr> "Municipality", "Municipality", "Municipality~
## $ Maximum_Daily_Withdrawal <dbl> 21.95, 23.22, 25.49, 17.70, 26.19, 23.60, 20.~
## $ Year                 <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 202~
```

```

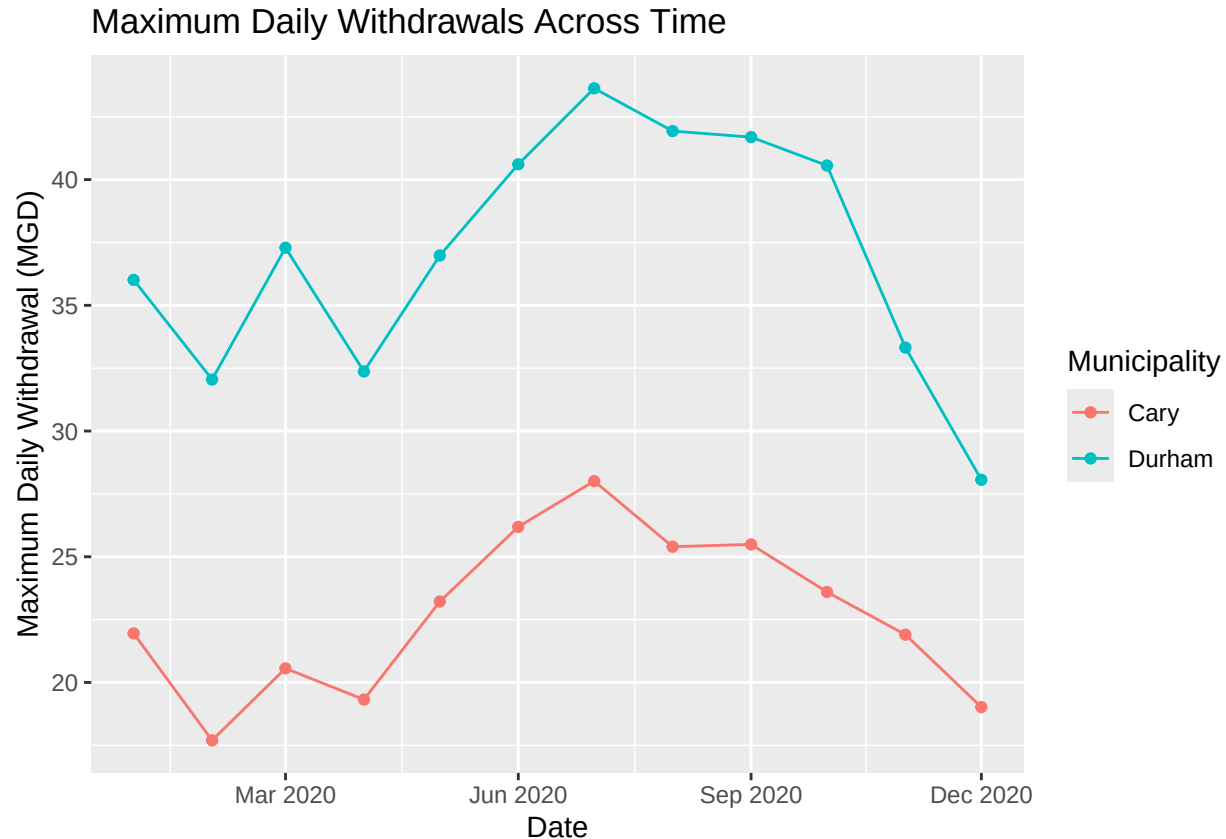
#change MGD column to numeric because it caused problems
dfC2020$Maximum_Daily-Withdrawal <-
  as.numeric(gsub(",", "", dfC2020$Maximum_Daily-Withdrawal))

#combine with Durham data collected above in a plot
#first use rbind to combine the two dataframes
#drop date column from df2020 for binding
df2020 <- df2020 %>% select(-Date)
df_combined = rbind(df2020, dfC2020)
#convert to date format
df_combined$Date <- as.Date(sprintf("%d-%02d-01",
                                     df_combined$Year, df_combined$Month))
df_withdrawals_2020_2 <-
  ggplot(df_combined, aes(x = Date, y = Maximum_Daily-Withdrawal,
                          group=Water_System_Name, color = Water_System_Name)) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  scale_x_date(date_labels = "%b %Y", date_breaks = "3 months") +
  #making sure months show on x axis since I got points off for that last hw
  labs(
    title = "Maximum Daily Withdrawals Across Time",
    x = "Date",
    y = "Maximum Daily Withdrawal (MGD)",
    color = "Municipality")

## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.

show(df_withdrawals_2020_2)

```

9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
#use function to extract data
#first create a vector of all years we will use
the_years = c(2018, 2019, 2020, 2021, 2022, 2023)
#use map2 to iteratively run the function and combine dataframes using bind_rows
dfs_Cary <- map2(the_years, "03-92-020", scrape.it) %>%
  bind_rows()
```

```
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-92-020&year=2018"
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-92-020&year=2019"
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-92-020&year=2020"
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-92-020&year=2021"
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-92-020&year=2022"
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-92-020&year=2023"
```

```

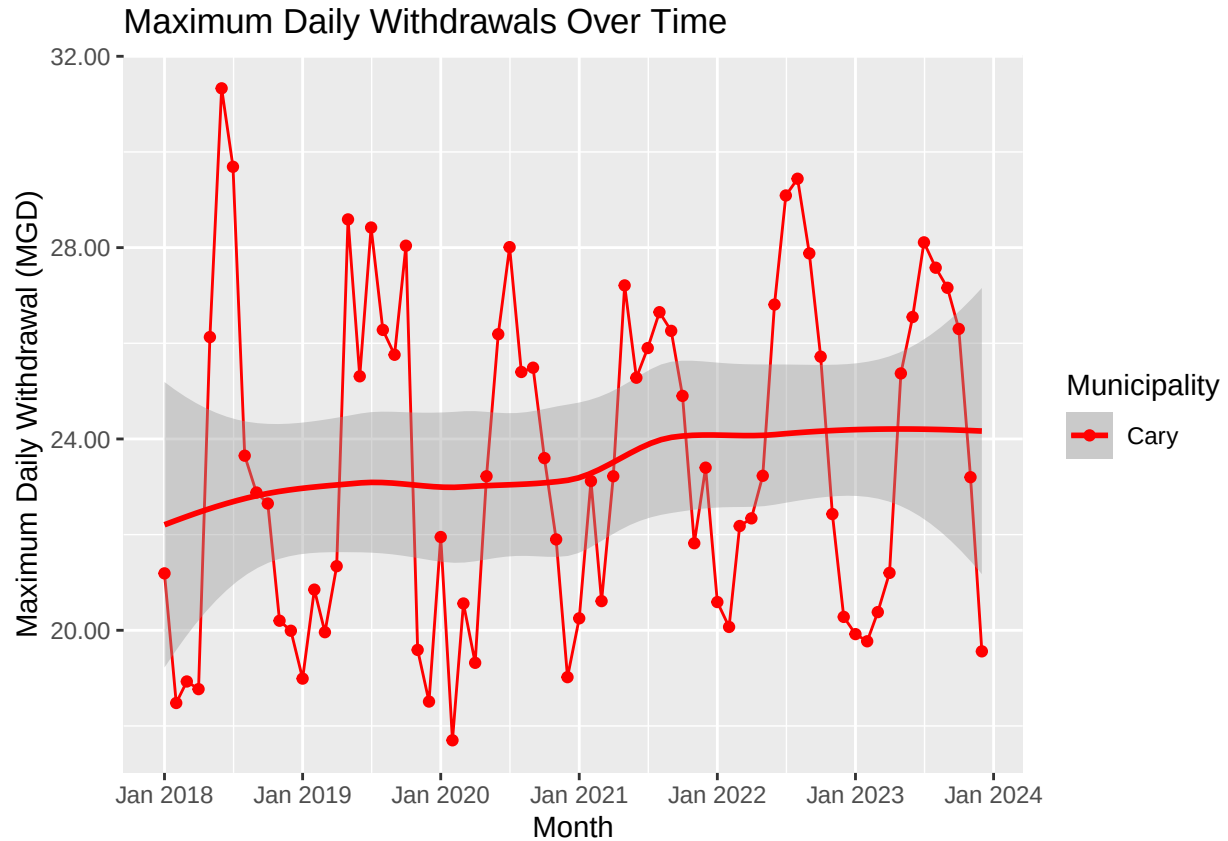
#show(dfs_Cary)
#add date column to dfs_Cary
dfs_Cary$Date <- as.Date(sprintf("%d-%02d-01", dfs_Cary$Year, dfs_Cary$Month))
#fix maximum daily withdrawal column
dfs_Cary$Maximum_Daily-Withdrawal <-
  as.numeric(gsub(",", "", dfs_Cary$Maximum_Daily-Withdrawal))

#create plot, similar to same code above
#plot edited from v1 to show one line for each source spanning years
dfC_plot <-
  ggplot(dfs_Cary, aes(x = Date, y = Maximum_Daily-Withdrawal,
                      group=Water_System_Name,
                      color = Water_System_Name)) +

  geom_line() +
  geom_smooth(method = "loess") +
  geom_point() + #add rainbow color palette for clarity
  scale_color_manual(values = rainbow(length(unique(dfs_Cary$Year)))) +
  scale_x_date(date_labels = "%b %Y", date_breaks = "1 year") +
  scale_y_continuous(labels = scales::label_number(accuracy = 0.01)) +
  #making sure months show on x axis since I got points off for that last hw
  labs(
    title = "Maximum Daily Withdrawals Over Time",
    x = "Month",
    y = "Maximum Daily Withdrawal (MGD)",
    color = "Municipality")
show(dfC_plot)

## 'geom_smooth()' using formula = 'y ~ x'

```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Cary's water usage follows a similar year over year pattern. It is lowest from January through March, in roughly the 17.5-22.5 MGD range. By March it begins steadily increasing to a summer peak of 25-30 MGD, around June through August. At which point it begins a steady decrease again until December. In December it reaches the same 17.5-22.5MGD range of values. Overall water usage does not seem to exhibit a trend from one year to the next. If anything, it slightly increased.